Python list comprehensions

A list comprehension builds a new list from an existing one in a single line. The pattern is [expression for item in iterable if condition]. For example, [n * n for n in range(10) if n % 2 == 0] gives the squares of the even numbers below ten.

They replace a loop that appends to an empty list, and they are usually easier to read when the logic is short. If the expression gets long or needs several conditions, a regular for loop is clearer.

Dictionary and set comprehensions work the same way with curly braces. A generator expression uses parentheses and produces values one at a time, which saves memory on large data.